Luciano Moffatt
Instituto de Química Física de los Materiales, Medio Ambiente y Energía (INQUIMAE), CONICET
Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires
Ciudad de Buenos Aires, C1428EHA, Argentina
lmoffatt@qi.fcen.uba.ar

30 August 2026

Dear Editors,

I am submitting "Likelihood approximations distort the ion channel kinetic information in macroscopic currents" for consideration as a Research Article.

Macroscopic currents have reached the limit of what they show by eye. The flip state we proposed for $P2X_2$ in 2007 was still on the visible side, a delay before the current rises. The questions now asked, whether its subunits couple in sequence or all at once, leave no such mark. They are beginning to be settled by comparing the Bayesian evidence for rival kinetic schemes, and how much the likelihoods behind such a comparison misreport the information a recording holds, and in which directions, had never been measured for macroscopic currents. This manuscript measures both.

Every likelihood that ignores the correlation between successive samples misreports its information by an order of magnitude, two- to threefold on the error bar, wherever that correlation is present in the record, and no single rescaling repairs it, because the distortion differs between parameter directions. The filter that uses the channel state at both ends of each sampling interval is calibrated: it reports the uncertainty it delivers, except where single openings become resolvable, and there its departure follows one measured power law.

I know the stake from the inside. In our 2025 ranking of nine $P2X_2$ schemes the winning mechanism changed when the likelihood's recursion was removed, overturning a factor of six in evidence; we published the sensitivity and argued for one of the two mechanisms. On a recording there is no truth to measure a likelihood against, and fitting well does not supply one.

So I went outside the fit. Channel gating simulates exactly where its likelihood cannot be evaluated exactly, and at known parameters a correct likelihood satisfies two identities: its score, the gradient of the log-likelihood, averages to zero, and the covariance of the score equals the Fisher information the likelihood itself reports. I measured both for eight likelihoods, from least squares to the filter above, against ten thousand exact simulations at every combination of channel number, noise and sampling interval tested, 560 combinations for the filter and 210 for all eight, in a two-state scheme, the smallest that isolates the approximation's own error.

Our control, the likelihood without the recursion, reports at one representative condition about twenty times the information the record holds; the likelihood we argued for is calibrated there. The 2025 evidence is not recomputed, and whether this carries to nine schemes is suggested and not shown.

The closest prior check on a filter of this kind appeared in eLife, by Münch and colleagues in 2022. They counted how often the true value fell inside the reported interval, at a fit per simulation, which covers a line of conditions. The two identities cost one likelihood evaluation per simulation and return a matrix: which parameter is misreported, and whether at each sample or across the record. The filter measured here was published with our 2025 $P2X_2$ analysis and has the form of a 1988 integrated-measurement Kalman filter; what is new is its score and Fisher information, and the measurement built on them.

For an experimentalist the map has one line. No measured condition gives both a calibrated least-squares error bar and information about the single-channel current; the two lie a factor of a hundred apart in noise, and still a factor of three under the paper's looser criterion. The recordings rich enough to determine the amplitudes are the ones whose least-squares error bars are the wrong width.

The engine with its exact simulator, configurations and notebooks (10.5281/zenodo.22168409), the simulation output (10.5281/zenodo.22167744) and the likelihood with its score, Fisher information and diagnostic, a small library with R and Python bindings (10.5281/zenodo.22168263), are public, so a reader can put the same two identities on their own scheme. No related manuscript is under consideration elsewhere.

Yours sincerely,

Luciano Moffatt